

# ERP System Implementation Project (Odoo)

Watan Designs Limited, Kingdom of Saudi Arabia

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Training Manual & Implementation Guide by:

Dubai in Cairo Ltd.

For Digital Marketing and eCommerce Services

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## Odoo Manufacturing Module (Chapter 03)

Release date: November 2025

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### Project Manager

Eng. Abdalla Hassan Elfouly

### Configuration Team

Mr. Ibrahim Mohamed Saad

Mr. Moataz Mohamed Mostafa

### Production Consultant

Eng. Amr Abdel-Mottaleb

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## Chapter 03: Manufacturing Resource Planning (MRP)

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### 3.1 Introduction and Objectives

Odoo's Manufacturing Resource Planning is a pivotal module that connects the sales, inventory, and purchasing departments. It aims to convert sales orders into actual production orders and to accurately monitor the workflow in workshops and production lines.

#### Module Objectives:

- Organizing the process of converting raw materials into finished products according to approved production orders.
- Regulating the relationship between Bills of Materials, work centers, and production lines.
- Track the manufacturing process step by step through work orders (WOs).
- Ensuring complete integration between inventory, manufacturing and sales so that materials are automatically deducted when production begins, and the finished product is added upon completion.
- Reducing waste and improving operational efficiency through performance monitoring and tracking production indicators.

### 3.2 Basic Terminology

| Term                              | Translation                    | Practical explanation                                                                                                                                        |
|-----------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bill of Materials<br>(BOMs)       | قائمة المواد                   | List of materials that make up the product, including all the raw materials and elements required.                                                           |
| Work Center<br>(WC)               | مركز عمل<br>أو نقطة إنتاج      | The machine, table, or point responsible for performing a specific operation within the production line.                                                     |
| Routing<br>(Process Flow)         | مسار التصنيع<br>(تسلسل التدفق) | The manufacturing process that defines the sequence of operations, from the first to the last work center, to complete the production of a specific product. |
| Work Order<br>(WO)                | أمر الشغل                      | A separate work order issued from the manufacturing order for each operation in the path.                                                                    |
| Manufacturing Order<br>(MO)       | أمر التصنيع                    | The basic manufacturing order that brings together all the components of the process from materials to final outputs.                                        |
| Operation                         | عملية تشغيل                    | The process which is carried out within the work center, such as cutting, assembling, or painting.                                                           |
| Scrap<br>(Waste)                  | مواد فهدرة                     | Any damage or loss during the manufacturing process, and it will be recorded in the system.                                                                  |
| By-Product<br>(Secondary Product) | منتج غير مقصود                 | A product resulting from the manufacturing process, such as sawdust produced from the cutting process.                                                       |

### 3.3 The Work Cycle Within The Manufacturing System

| Stage | Goal                                            | Brief Description                                                                                        |
|-------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1     | Preparing A List Of Materials                   | Define the raw materials that make up each product, specifying the required quantities, to make the BOM. |
| 2     | Path Preparation                                | Identifying the sequence of operations and work centers used in manufacturing.                           |
| 3     | Create A Manufacturing Order (MO)               | Create a production order based on a sales request coming from one of the sales channels.                |
| 4     | Issuing A Work Order (WO)                       | Create automatic work orders for each process in the specified path.                                     |
| 5     | Dispensing Raw Materials For Production         | Raw materials are deducted from inventory and delivered to the manufacturing department.                 |
| 6     | Implementing Operations And Monitoring Progress | Track the status of each process (running/completed).                                                    |
| 7     | Recording Losses And Additions                  | Record any damaged materials or by-products at "ShopFloor".                                              |
| 8     | Receiving The Final Product                     | Add the finished product to the inventory after the manufacturing order has been completed.              |

### 3.4 System Operating Procedures (Step-by-Step)

#### A. Creating a Bill of Materials

1. From the main menu, select Manufacturing → Products → Bill of Material (BOM).
2. Click “Create” and then select the final product.
3. Add all the required raw materials and their quantities.
4. Specify the routing if there is more than one production process.
5. Press Save to approve the list.

*Note: It is normal to create more than one list of materials for the same product as long as there are different specifications.*

#### B. Creating the routing and defining work centers

1. From the menu: Manufacturing → Settings → Routes.
2. Click “Create” and add the path name.
3. Add the operations in order (cutting - assembling - painting - filling).
4. For each operation, specify the Work Center and the estimated duration.
5. Save the path to link it later to the list of materials.

#### C. Creating a Manufacturing Order (MO)

1. From the menu: Manufacturing → Manufacturing Orders.
2. Click Create, then select the product you want to manufacture.
3. The system will automatically load the list of materials and the associated path.
4. Click “Confirm” to change it to the “In Production” status.
5. The resulting work orders can be displayed from the same screen.

#### D. Executing Work Orders

1. From the "Work Orders" tab within the manufacturing command, select the first operation and press "Start".
2. When finished, press "Mark as Done" to update the status.
3. Repeat the step for each operation in the path until all stages are completed.
4. Notes can be added, or an actual time can be recorded that differs from the estimated time.
5. In case of a malfunction or interruption, the reason can be recorded on the same page.

*Note: You must press Pause if the machine or operator stops performing the task for any reason.*

#### E. Raw material expenditure for production

1. The system automatically deducts materials when a manufacturing order is started, but they can be reviewed manually from the raw materials tab.
2. In case of shortage of material, a purchase order (PO) or internal transfer can be created directly from the screen.
3. After confirmation, the quantities are automatically deducted from the inventory and sent to the production site.

#### F. Receiving the final product

1. After all work orders have been completed, press "Mark as Done" to finish production.
2. Enter the final quantity produced, with the possibility of adjusting for losses.
3. The system automatically creates a manufacturing receipt.
4. After approval, the finished product is added to the inventory and the manufacturing order (MO) is closed.

### 3.5 Key Performance Indicators (KPIs)

| Index                                                 | Description                                                                              | Ref. Goal   |
|-------------------------------------------------------|------------------------------------------------------------------------------------------|-------------|
| Production Efficiency<br>(كفاءة الإنتاج)              | Rate of adherence to planned production time compared in real time.                      | $\geq 90\%$ |
| Material Consumption Accuracy<br>(دقة استهلاك المواد) | Accuracy of material consumption compared to specified quantities pre-listed in the BOM. | $\geq 95\%$ |
| Downtime Rate<br>(معدل التوقف)                        | The rate of business center closures.                                                    | $\leq 5\%$  |
| Rework Ratio<br>(نسبة إعادة التشغيل)                  | The percentage of orders that required remanufacturing or modification.                  | $\leq 3\%$  |
| On-Time Delivery<br>(معدل التسليم المنتظم)            | Percentage of production orders completed on schedule.                                   | $\geq 90\%$ |

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### 3.6 Frequently Asked Questions (FAQ)

Q: What is the difference between a Manufacturing Order (MO) and Work Orders?

A: A manufacturing order is the complete process for the final product, while work orders are the partial steps carried out at different work centers within the same order.

Q: Can the list of materials be modified after it has been approved?

A: The list of materials can be copied and a new, modified version created, but it is not recommended to modify the approved list after it has been used in actual manufacturing orders.

Q: How is scrap recorded?

A: From within the work order or manufacturing order, you can click on "Record Waste" and specify the quantity and reason, and it will be deducted from the inventory automatically.

Q: What is the benefit of routing?

A: It determines the order of operations required to produce the product, and helps in tracking time and stages accurately and linking each operation to the work center responsible for it.

Q: Can a manufacturing order (MO) be directly linked to a sales order?

A: Yes, in the case of the Made-to-Order model, an automatic manufacturing order is created for each approved sales order.

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### 3.7 Data Entry Rules (Data Hygiene)

- Ensure that lists of materials, centers, and routes are accurately defined before production.
- Update the actual runtime after each operation to correct efficiency data.
- Introduce waste and by-products immediately after they appear.
- Review the status of each manufacturing order daily to ensure proper closure.
- Avoid repeating the definition of the same material under different names.
- Ensuring that each product is linked to the correct material list and the correct route.
- Review material consumption reports weekly to adjust for deviations.

### 3.8 Production Supervisor Daily Checklist

1. Review the list of open manufacturing orders at the beginning of the day.
2. Ensure that raw materials are available in stock before starting operations.
3. Monitoring each operation on the system in real time and recording any downtime.
4. Record the quantities produced and wasted after each operation.
5. Approving completed work orders and updating the status on the system.
6. Delivering the finished products to the inventory after final review.
7. Checking the daily performance indicators for work centers.